

Basic Concepts

1. Class and Object Creation

- Create a Book class with attributes: title, author, and price. Write a main method to create three Book objects and display their details.

2. Encapsulation

- Create a BankAccount class with private fields accountNumber, accountHolderName, and balance. Implement getter and setter methods to access and update the balance safely.

3. Method Overloading

- Create a Calculator class with overloaded methods add(int a, int b), add(int a, int b, int c), and add(double a, double b). Write a main method to test all three methods.

Intermediate Concepts

4. Constructor Overloading

- Create a Student class with two constructors:
 - One that accepts name and rollNumber.
 - Another that accepts name, rollNumber, and grade.
- Test both constructors by creating appropriate objects in the main method.

5. Static Members

- Create a Company class with a static field companyName and a non-static field employeeName. Write a static method to update companyName. Demonstrate its usage by creating multiple objects.

6. Association (Composition)

- Create a Library class that contains an array of Book objects. Write methods in Library to add books and display the details of all books.

Advanced Concepts

7. Polymorphism

- Create an Animal class with a method makeSound(). Derive Dog and Cat classes from Animal and override the makeSound() method in each. Use dynamic method dispatch to call the appropriate method.

8. Abstraction

- Create an abstract class Shape with an abstract method calculateArea(). Implement Circle and Rectangle classes that extend Shape and provide specific implementations for calculateArea(). Test them in the main method.

9. Interface

- Define an interface Playable with a method play(). Implement it in two classes: Football and VideoGame. Create objects of each class and call their play() method.

10. File Operations (Combining OOP with IO)

- Create a Contact class with attributes name and phoneNumber. Write a program to:
 - Accept multiple contacts from the user.
 - Save the contacts to a file.
 - Read the contacts back from the file and display them.